

Real-Time Intelligence and Scalable Architecture for Robotics

Title page and declarations
(not included in the blind-reviewed PDF)

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Declarations

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Competing interests

The authors declare that they have no competing interests.

Author contributions

S.M. led the conceptualization and writing. D.C. contributed systems-architecture and reliability perspectives. G.R.N. contributed survey synthesis and safety/assurance perspectives. All authors reviewed and approved the final manuscript.

Data availability

Simulation outputs supporting the discrete-event evaluation are included as CSV/JSON artifacts (`artifacts/des_outputs/`, including per-run summaries, CAAI sensitivity sweep, and `des_plot_samples.csv`). A host-side timing trace is provided as `artifacts/host_trace_latency_ms.csv`. No proprietary or human-subject datasets were used.

Code availability

A software-only reference implementation of a TIC monitor and CAAI supervisor is available at <https://github.com/sanmish4ds/robot-scalable-architect> (directory `tic_monitor/`), with JSON Schema at `tic_monitor/schema/tic.schema.json`.

Ethics approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.